

Eight families, four evidence questions

Completion is stated per row; proximity is not proof transfer.



analytic theorem / exact analysis



kernel- or solver-checked subclaim



bounded corpus / replay / fixture



document or integration state

RESULT FAMILY exact scope named in every row	ANALYTIC what is proved on paper?	FORMAL which subclaims are checked?	BOUNDED EXECUTABLE what finite evidence ran?	REPORT / INTEGRATION not a mathematical proof level
1 Foundational categorical Sx audit	Exact semantics + counterexamples	Lean semantic factorization slice	Exact witnesses + bounded checks	SCOPED DETAILED REPORT Markdown + PDF
2 Finite-alphabet plug-in convergence	Exact-real convergence theorem	Lean deterministic continuity core	Finite fixtures + Rust comparison	SCOPED DETAILED REPORT Markdown + PDF
3 Support-change-tolerant averaged Sx continuity	Exact-real continuity moduli	Lean algebraic subclaims	18 law pairs + bounded replay	SCOPED DETAILED REPORT Markdown + PDF
4 Dependency-color concentration and local bounds	Probability theorem + local Sx bounds	Lean deterministic subclaims only	Oracle + Rust bounded checks	SCOPED DETAILED REPORT Markdown + PDF
5 Exact two-source Sx assurance	Count → atoms + product theorem	Lean bridge + product algebra	Exact certifier + bounded routes	THREE DISTINCT LAYERS Multiple scoped PDFs Certifier conditional Integration open
SxPID3 source-marginal factorization and bounded audit	Source-marginal factorization	Lean generic finite-event theorem	20,348 tables two exact routes	INTEGRATED • SCOPED Markdown + PDF Full certificate remains open
Represented-binary64 and quantizer assurance	Exact represented operand reduction	No end-to-end formal proof claimed	Exact oracles + bounded tests	CANONICAL MARKDOWN Dedicated PDF open
KSG integer-harmonic local arithmetic	Harmonic identity + sharp box bound	Lean + Z3 scoped arithmetic	8,198-row bounded corpus	SCOPED CORE: GO Exact • formal • bounded Integration: NO-GO Dedicated result PDF open

READ THE CELLS LITERALLY

Formal subclaims, exact arithmetic, bounded corpora, report PDFs, and integration gates remain separate evidence objects.